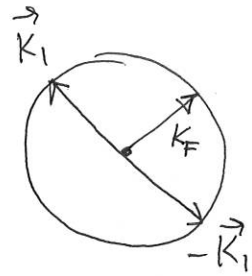
Cooper problem and spin structure of cooper pairs.



2-electron problem (Cooper Phys. Rev. 104 p. 1189 (1956))

$$-\frac{\hbar^2}{2m}\left(\vec{\nabla}_1^2+\vec{\nabla}_2^2\right)\psi(\vec{r}_1,\vec{r}_2)+V(\vec{r}_1-\vec{r}_2)\,\psi(\vec{r}_1,\vec{r}_2)=\left(\Delta+2\cdot\frac{\hbar^2K_F^2}{2m}\right)\psi(\vec{r}_1,\vec{r}_2)$$

energy of the pair

$$\vec{R}=\frac{1}{2}(\vec{r}_1+\vec{r}_2)\,,\quad \vec{r}=\vec{r}_1-\vec{r}_2 \Rightarrow \psi(\vec{r}_1,\vec{r}_2)=\underbrace{e^{i\vec{K}\cdot\vec{R}}}_{\text{w.f. of CM motion}}\psi(\vec{r})$$

$$\Rightarrow -\frac{\hbar^2}{m}\vec{\nabla}_r\psi(\vec{r})+V(\vec{r})\psi(\vec{r})=\left(\Delta^2+\frac{\hbar^2K_F^2}{m}\right)\psi(\vec{r})$$ (Assume $\vec{K}=0$)

let $g(\vec{k})=\int d\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\,\psi(\vec{r})$

$$\frac{\hbar^2k^2}{m}g(\vec{k})+\int\frac{d\vec{k}'}{(2\pi)^3}\,V(\vec{k}-\vec{k}')\,g(\vec{k}')=\left(\Delta+\frac{\hbar^2K_F^2}{m}\right)g(\vec{k})$$

Interaction $V(\vec{k}-\vec{k}')=\int d^3\vec{r}\; e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}}\,V(\vec{r})$

depends on momenta $\vec{k},\vec{k}'$ mostly via direction (i.e. $|\vec{k}|=|\vec{k}'|=K_F$).

Expand in angular harmonics

$$V(\vec{k}-\vec{k}')=\sum_{l=0}^{\infty}V_l(k,k')\sum_{m=-l}^{l}Y_{lm}(\hat{k})\,Y^*_{lm}(\hat{k}')$$

$Y_{lm}(\hat{k})$ = spherical function with orbital momentum l, projection m, $\hat{k}=\frac{\vec{k}}{K_F}$ unit vector on FS.

$$\int\frac{d\Omega}{4\pi}\,Y_{lm}(\Omega)\,Y^*_{l'm'}(\Omega)=\delta_{m,m'}\,\delta_{l,l'}$$

$d\Omega = d(\cos\theta)\,d\varphi$

Details

$$\int d\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\; \vec{\nabla}_r^2\, \psi(\vec{r}) = \int d\vec{r}\; \frac{\partial}{\partial \vec{r}}\left(e^{-i\vec{k}\cdot\vec{r}}\, \frac{\partial}{\partial \vec{r}}\, \psi(\vec{r})\right) -$$

$$- \int d\vec{r}\; \frac{\partial}{\partial r}\left(e^{-i\vec{k}\cdot\vec{r}}\right) \frac{\partial}{\partial \vec{r}}\, \psi(\vec{r}) = (-)\int d\vec{r}\; \frac{\partial}{\partial \vec{r}}\left\{ \frac{\partial}{\partial \vec{r}}\left(e^{-i\vec{k}\cdot\vec{r}}\right)\psi(\vec{r})\right\}$$

$$+ \int d\vec{r}\; \frac{\partial^2}{\partial \vec{r}^{\,2}}\left(e^{-i\vec{k}\cdot\vec{r}}\right)\psi(\vec{r}) = -\vec{k}^2 \int d\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\, \psi(\vec{r}) =$$

$$= -\vec{k}^2\, g(\vec{k}).$$

$$\int d\vec{r}\; V(\vec{r})\, \psi(\vec{r})\, e^{-i\vec{k}\cdot\vec{r}} \Rightarrow$$

$$V(\vec{r}) = \int \frac{d\vec{q}}{(2\pi)^3}\, e^{i\vec{q}\cdot\vec{r}}\, V(\vec{q})$$

$$\psi(\vec{r}) = \int \frac{d\vec{k}'}{(2\pi)^3}\, e^{i\vec{k}'\cdot\vec{r}}\, g(\vec{k}')$$

$$\Rightarrow \int d\vec{r} \int \frac{d\vec{q}}{(2\pi)^3} \int \frac{d\vec{k}'}{(2\pi)^3}\, e^{i\vec{q}\cdot\vec{r} + i\vec{k}'\cdot\vec{r} - i\vec{k}\cdot\vec{r}}\, V(\vec{q})\, g(\vec{k}') =$$

$$\int d\vec{r}\; e^{i\vec{r}\cdot(\vec{q}+\vec{k}'-\vec{k})} = (2\pi)^3\, \delta(\vec{q}+\vec{k}'-\vec{k})$$

$$= \int \frac{d\vec{k}'}{(2\pi)^3}\, V(\vec{k}-\vec{k}')\, g(\vec{k}')$$

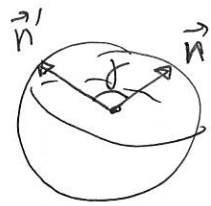
Usual expansion $f(\theta,\varphi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} C_{\ell m}\, Y_{\ell m}(\theta,\varphi)$.

But we use the <u>addition theorem</u>

$$P_\ell(\cos\gamma) = P_\ell(\cos\theta)\, P_\ell(\cos\theta') + \sum_{m=1}^{\ell} 2\, \frac{(\ell-m)!}{(\ell+m)!}\, P_\ell^m(\cos\theta)\, P_\ell^m(\cos\theta') \cdot \cos(m(\varphi-\varphi'))$$

where $P_\ell^m(\cos\theta)$ = Legendre polynomials, $P_\ell^{m=0} = P_\ell$,

and



$$\vec{n}\cdot\vec{n}' = \cos\gamma = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi-\varphi')$$

$$\vec{n} = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)$$

$$\vec{n}' = (\sin\theta'\cos\varphi',\ \sin\theta'\sin\varphi',\ \cos\theta')$$

Then $P_\ell(\vec{n}\cdot\vec{n}') = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\vec{n}')\, Y_{\ell m}(\vec{n})$

where $Y_{\ell m}(\theta,\varphi) = (-1)^{\frac{m+|m|}{2}}\, i^\ell \left[\frac{2\ell+1}{4\pi}\, \frac{(\ell-|m|)!}{(\ell+|m|)!}\right]^{1/2} P_\ell^{|m|}(\cos\theta)\, e^{im\varphi}$

So that $V(\vec{k}-\vec{k}') = \sum_{\ell=0}^{\infty} V_\ell(k,k') \sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{k})\, Y_{\ell m}^*(\hat{k}')$

represents expansion of $V(\vec{k}-\vec{k}')$ in relative angle γ,

where $|\vec{k}-\vec{k}'|^2 = k^2 + k'^2 - 2kk'\cos\gamma$.

Thus expansion in $Y_{\ell m}$ is the same as expansion in $P_\ell(\cos\gamma)$

$$V(\vec{k}-\vec{k}') = \sum_\ell (2\ell+1)\, V_\ell(k,k')\, P_\ell(\cos\gamma)$$

but is more convenient for later purposes.

$P_{00}(\cos\gamma) = 1$; $P_{10}(\cos\gamma) = \cos\gamma$, $P_{11} = -\sin\gamma$; $P_{20} = \frac{1}{2}(3\cos^2\gamma - 1)$

$P_{21} = -3\sin\gamma\cos\gamma$, $P_{22} = 3\sin^2\gamma$ ...

Let us now <u>assume</u>

$$V_\ell(k,k') = \begin{cases} -V_\ell & \varepsilon_F \le \frac{\hbar^2 k^2}{2m}, \frac{\hbar^2 k'^2}{2m} \le \varepsilon_F + \varepsilon_\ell \\ 0 & \text{otherwise} \end{cases}$$

Integrate using $\int \frac{d^3\vec{k}}{(2\pi)^3} \to \nu \int d\xi \int \frac{d\Omega}{4\pi}$, $\nu = \frac{m k_F}{2\pi^2 \hbar^2}$

$$\Rightarrow \left[\frac{\hbar^2}{m}(k^2 - k_F^2) - \Delta\right] g(\vec{k}) = \nu \int_0^{\varepsilon_\ell} d\xi' \int \frac{d\Omega'}{4\pi} \sum_{\ell=0}^{\infty} V_\ell \sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{k}) Y^*_{\ell m}(\hat{k}') \cdot g(\vec{k}')$$

let $g(\vec{k})$ be a state with definite angular momentum ℓ': $g(\vec{k}) = g_{\ell'}(\vec{k}) = \sum_{m'=-\ell'}^{\ell'} a_{\ell' m'}(k) \cdot Y_{\ell' m'}(\hat{k})$

$\Rightarrow$ then collect terms that depend on $\hat{k}'$:

and use $\int \frac{d\Omega'}{4\pi} Y^*_{\ell m}(\hat{k}') Y_{\ell' m'}(\hat{k}') = \delta_{\ell\ell'} \delta_{mm'}$

$$\Big[\underbrace{\frac{\hbar^2}{m}(k^2 - k_F^2)}_{2\xi} - \Delta_{\ell'}\Big] g_{\ell'}(\hat{k}) = \nu V_{\ell'} \int_0^{\varepsilon_{\ell'}} d\xi\, g_{\ell'}(\vec{k})$$

$$\Rightarrow g_\ell(\vec{k}) = \frac{\nu V_\ell}{2\xi - \Delta_\ell} \int_0^{\varepsilon_\ell} d\xi\, g_\ell(\vec{k})$$

Divide by $\int_0^{\varepsilon_\ell} d\xi\, g_\ell(\vec{k}) \neq 0 \Rightarrow 1 = \nu V_\ell \int_0^{\varepsilon_\ell} \frac{d\xi}{2\xi - \Delta_\ell} =$

$$= \frac{1}{2} \nu V_\ell \ln \frac{\varepsilon_\ell - \frac{1}{2}\Delta_\ell}{-\frac{1}{2}\Delta_\ell}$$

$$e^{\frac{2}{\nu V_\ell}} = \frac{\varepsilon_\ell - \frac{1}{2}\Delta_\ell}{-\frac{1}{2}\Delta_\ell} \Rightarrow \frac{1}{2}\Delta_\ell\left(1 - e^{\frac{2}{\nu V_\ell}}\right) = \varepsilon_\ell$$

$$\Delta_\ell = \frac{2\varepsilon_\ell}{1 - e^{\frac{2}{\nu V_\ell}}} = -2\varepsilon_\ell e^{-\frac{2}{\nu V_\ell}}.$$

This is binding energy ($\Delta_\ell < 0$) of two electrons with orbital w.f.

$$g_\ell(\vec{k}) = \sum_{m=-\ell}^{\ell} a_{\ell m}(k)\, Y_{\ell m}(\hat{k}).$$

But $g_\ell(-\vec{k}) = (-1)^\ell g_\ell(\vec{k})$ and full w.f. $\overbrace{g(\vec{k})\chi_{12}}$ must be antisymmetric: $g(-\vec{k})\chi_{21} = -g(\vec{k})\chi_{12}$

$\Rightarrow$ ℓ = even: $\chi_{21} = -\chi_{12}$ antisymmetric spin state $\Rightarrow$ spin singlet

ℓ = odd: $\chi_{21} = \chi_{12}$ symmetric spin state

Spin basis: $\alpha_\lambda = |\uparrow\rangle = \begin{pmatrix}1\\0\end{pmatrix}$

$\beta_\lambda = |\downarrow\rangle = \begin{pmatrix}0\\1\end{pmatrix}$

<u>Singlet</u> $|\uparrow\downarrow - \downarrow\uparrow\rangle = \underset{1}{\alpha_\lambda}\,\underset{2}{\beta_\mu} - \underset{1}{\beta_\lambda}\,\underset{2}{\alpha_\lambda}$

This is direct product of two spinors

$$\vec{\alpha} \otimes \vec{\beta} = \{\alpha_\lambda \beta_\mu\} = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}^T - \begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix}^T$$

$$\begin{pmatrix}0 & 1\\0 & 0\end{pmatrix} - \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix}$$

$$= \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix} = i\hat{\sigma}_y.$$

$$\Rightarrow \psi_{pair}^{l=0,2,4\ldots} = g(\vec{k})\, i\sigma_y = \sum_{m=-l}^{l} a_{lm}\, Y_{lm}(\vec{k})\, i\sigma_y$$

<u>Triplet</u>

$$|\uparrow\uparrow\rangle = \alpha_\lambda \alpha_\mu = \begin{pmatrix}1\\0\end{pmatrix}\otimes\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1&0\\0&0\end{pmatrix} \qquad \hat{S}^z_{TOT}\ 1$$

$$|\uparrow\downarrow+\downarrow\uparrow\rangle = \begin{pmatrix}0&1\\1&0\end{pmatrix} \qquad 0$$

$$|\downarrow\downarrow\rangle = \beta_\lambda \beta_\mu = \begin{pmatrix}0&0\\0&1\end{pmatrix} \qquad -1$$

$$\psi_{pair}^{l=1,3,\ldots} = g_1(\vec{k})|\uparrow\uparrow\rangle + g_0(\vec{k})|\uparrow\downarrow+\downarrow\uparrow\rangle + g_{-1}(\vec{k})|\downarrow\downarrow\rangle$$

$$= \begin{pmatrix} g_1 & g_0 \\ g_0 & g_{-1}\end{pmatrix}$$

where $g_s(\vec{k})$ is orbital w.f. of a pair with total spin $S = 1, 0, -1$ correspondingly:

$$g_s(\vec{k}) = \sum_{m=-l}^{l} a^s_{lm}\, Y_{lm}(\vec{k});$$

let us write $\psi_{pair}^{l=odd} = \left(\vec{d}(\vec{k})\cdot\vec{\sigma}\right)\left(i\hat{\sigma}^y\right) =$

$$= \left(d_x(\vec{k})\,\hat{\sigma}^x + d_y(\vec{k})\,\hat{\sigma}^y + d_z(\vec{k})\,\hat{\sigma}^z\right)\left(i\hat{\sigma}^y\right)$$

$$= d_x\, i\underbrace{\hat{\sigma}^x\hat{\sigma}^y}_{i\sigma^z} + d_y\, i\underbrace{\hat{\sigma}^y\hat{\sigma}^y}_{\hat{\sigma}^0} + d_z\, i\underbrace{\hat{\sigma}^z\hat{\sigma}^y}_{-i\hat{\sigma}^x} =$$

$$= -d_x\,\hat{\sigma}^z + i d_y\,\hat{\sigma}^0 + d_z\,\hat{\sigma}^x =$$

$$= \begin{pmatrix} -d_x + i d_y & d_z \\ d_z & d_x + i d_y\end{pmatrix}$$

For $l=1$, $m=-1,0,1$ and $\alpha = x,y,z \Rightarrow$ 9 complex functions b^α_{lm}

$$\Rightarrow \left.\begin{aligned} g_1 &= -d_x + i d_y \\ g_2 &= d_z,\ g_{-1} = d_x + i d_y \end{aligned}\right\} \Rightarrow d_{\alpha=x,y,z}(\vec{k}) = \sum_{m=-l}^{l} b^\alpha_{lm}\, Y_{lm}(\hat{k})$$

$$\langle \vec{S}_{el} \rangle = \frac{\hbar}{2} \langle \vec{\sigma} \rangle = \frac{\hbar}{2} \int \frac{d\Omega}{4\pi} \underline{Tr} \left(\psi^{l}_{pair} \right)^{*} \vec{\sigma} \left(\psi^{l}_{pair} \right) =$$

$$= \frac{\hbar}{2} \int \frac{d\Omega}{2\pi} \underline{Tr} \left[\left(d_x \sigma_x + d_y \sigma_y + d_z \sigma_z \right) i \sigma_y \right]^{*} \vec{\sigma} \left(d_x \sigma_x + d_y \sigma_y + d_z \sigma_z \right) \cdot i \sigma_y =$$

$$= \frac{\hbar}{2} \int \frac{d\Omega}{4\pi} \underline{Tr} - i \sigma_y \left(d_x^{*} \sigma_x + d_y^{*} \sigma_y + d_z^{*} \sigma_z \right) \vec{\sigma} \cdot$$

$$\cdot \left(d_x \sigma_x + d_y \sigma_y + d_z \sigma_z \right) i \sigma_y$$

$\sigma_a \sigma_b = \delta_{ab} + i \varepsilon^{abc} \sigma_c$ and $\underline{Tr}$ = matrix trace

Consider x-component: $\vec{\sigma} \to \sigma_x$

$$\langle S^{x}_{el} \rangle = \frac{\hbar}{2} \int \frac{d\Omega}{4\pi} \underline{Tr}\, \sigma_y \left(d_x^{*} \sigma_x + d_y^{*} \sigma_y + d_z^{*} \sigma_z \right) \left(d_x + d_y\, i\sigma_z + d_z (-i\sigma_y) \right) \sigma_y$$

$$= \frac{\hbar}{2} \int \frac{d\Omega}{4\pi} \underline{Tr} \left(d_x^{*} (-i\sigma_z) + d_y^{*} + d_z^{*}\, i \sigma_x \right) \left(d_x \sigma_y + i d_y (-i\sigma_x) + \right.$$

$$\left. + (-i d_z) \right) =$$

$$= \frac{\hbar}{2} \int \frac{d\Omega}{4\pi} \underline{Tr} \left(-i\, d_y^{*} d_z + i d_z^{*} d_y + \underline{\text{terms} \sim \sigma_\alpha} \right)$$

$$= \frac{i\hbar}{2} \int \frac{d\Omega}{4\pi} \left(d_z^{*} d_y - d_y^{*} d_z \right) \quad \text{since } \underline{Tr}(\sigma_\alpha) = 0.$$

This is found for S^{x}_{el} of one of electrons of the pair.

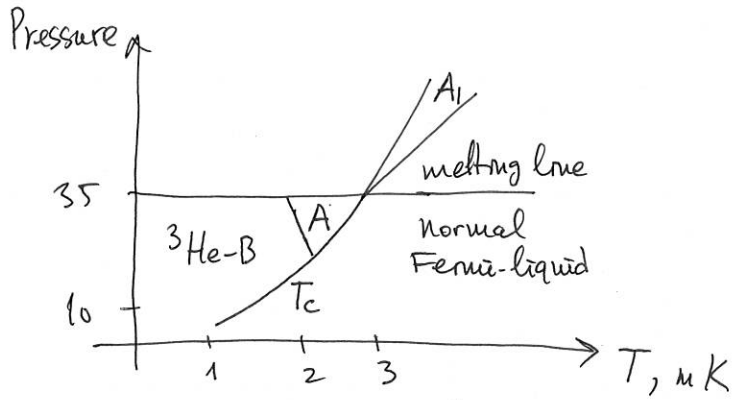
The total spin $\vec{S}_{TOT} = \vec{S}_1 + \vec{S}_2 = \frac{\hbar}{2} (\vec{\sigma}_1 + \vec{\sigma}_2)$

$$\Rightarrow \langle S^{x}_{TOT} \rangle = 2 \frac{\hbar}{2} \langle \sigma_1^{x} \rangle = i\hbar \int \frac{d\Omega}{4\pi} \underbrace{\left(d_z^{*} d_y - d_y^{*} d_z \right)}_{\text{x-component of the vector product.}}$$

$\Rightarrow$ can expect general result

$$\langle \vec{S}_{pair} \rangle = i\hbar \int \frac{d\Omega}{4\pi} \vec{d}^{*}(\vec{k}) \times \vec{d}(\vec{k})$$

Superfluid ^3He = p-wave superfluid



Superfluid with $\underline{l=1}$ $\Rightarrow$ $Y_{10} = \frac{1}{2}\sqrt{\frac{3}{\pi}}\cos\theta \sim \hat{k}_z$

$$Y_{11} = -\frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta\, e^{i\varphi} \sim -\frac{1}{\sqrt{2}}(\hat{k}_x + i\hat{k}_y)$$

$$Y_{1-1} = \frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta\, e^{-i\varphi} \sim \frac{1}{\sqrt{2}}(\hat{k}_x - i\hat{k}_y)$$

Then $d_s(\vec{k}) = \sum_{m=-l}^{l} a^s_{lm}\, Y_{lm}(\vec{k}) =$

$\underline{(s=1,0,-1)}$

$$= a^s_{10}\, Y_{10} + a^s_{11}\, Y_{11} + a^s_{1-1}\, Y_{1-1} =$$

$$= a^s_{10}\,\hat{k}_z - \frac{a^s_{11}}{\sqrt{2}}(\hat{k}_x + i\hat{k}_y) + \frac{a^s_{1-1}}{\sqrt{2}}(\hat{k}_x - i\hat{k}_y)$$

$$= \frac{-a^s_{11} + a^s_{1-1}}{\sqrt{2}}\,\hat{k}_x + \frac{-a^s_{11} - a^s_{1-1}}{\sqrt{2}}\, i\,\hat{k}_y + a^s_{10}\,\hat{k}_z$$

$\Rightarrow$ can write $d_s(\vec{k}) = \underbrace{A_{si}}_{3\times 3 \text{ matrix}}\,\hat{k}_i$ $\left(\begin{matrix} i = x,y,z \\ s = 1,0,-1 \end{matrix}\right)$

Thus matrix A_{si} plays the role of the order parameter.

$\underline{^3He\ B\text{-phase}}$: $\vec{d} \sim \hat{k}$ $\Rightarrow$ so that $A_{\alpha i} \sim \delta_{\alpha i}$ (unity matrix)

$$\psi^{B}_{pair} = \begin{pmatrix} -\hat{k}_x + i\hat{k}_y & \hat{k}_z \\ \hat{k}_z & \hat{k}_x + i\hat{k}_y \end{pmatrix} = (-\hat{k}_x + i\hat{k}_y)\,|\uparrow\uparrow\rangle + \hat{k}_z\,|\uparrow\downarrow + \downarrow\uparrow\rangle + (\hat{k}_x + i\hat{k}_y)\,|\downarrow\downarrow\rangle.$$

<u>A phase</u> $\vec{d} \sim (\hat{k}_x + i\hat{k}_y, 0, 0)$

$$\Psi^{A}_{pair} \sim (\hat{k}_x + i\hat{k}_y)\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -(\hat{k}_x + i\hat{k}_y)\left(|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle\right)$$

orbital momentum $L_z = 1$

<u>A_1 phase</u> $\vec{d} \sim (\hat{k}_x + i\hat{k}_y, i(\hat{k}_x + i\hat{k}_y), 0)$

$$\Psi^{A_1}_{pair} \sim (\hat{k}_x + i\hat{k}_y)\begin{pmatrix} -1 + i\cdot i & 0 \\ 0 & 1 + i\cdot i \end{pmatrix} =$$

$$= (\hat{k}_x + i\hat{k}_y)\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = (\hat{k}_x + i\hat{k}_y)\,|\uparrow\uparrow\rangle$$

$L_z = +1$ $\quad S^z_{TOT} = +1$